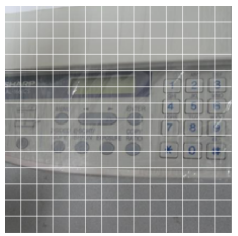
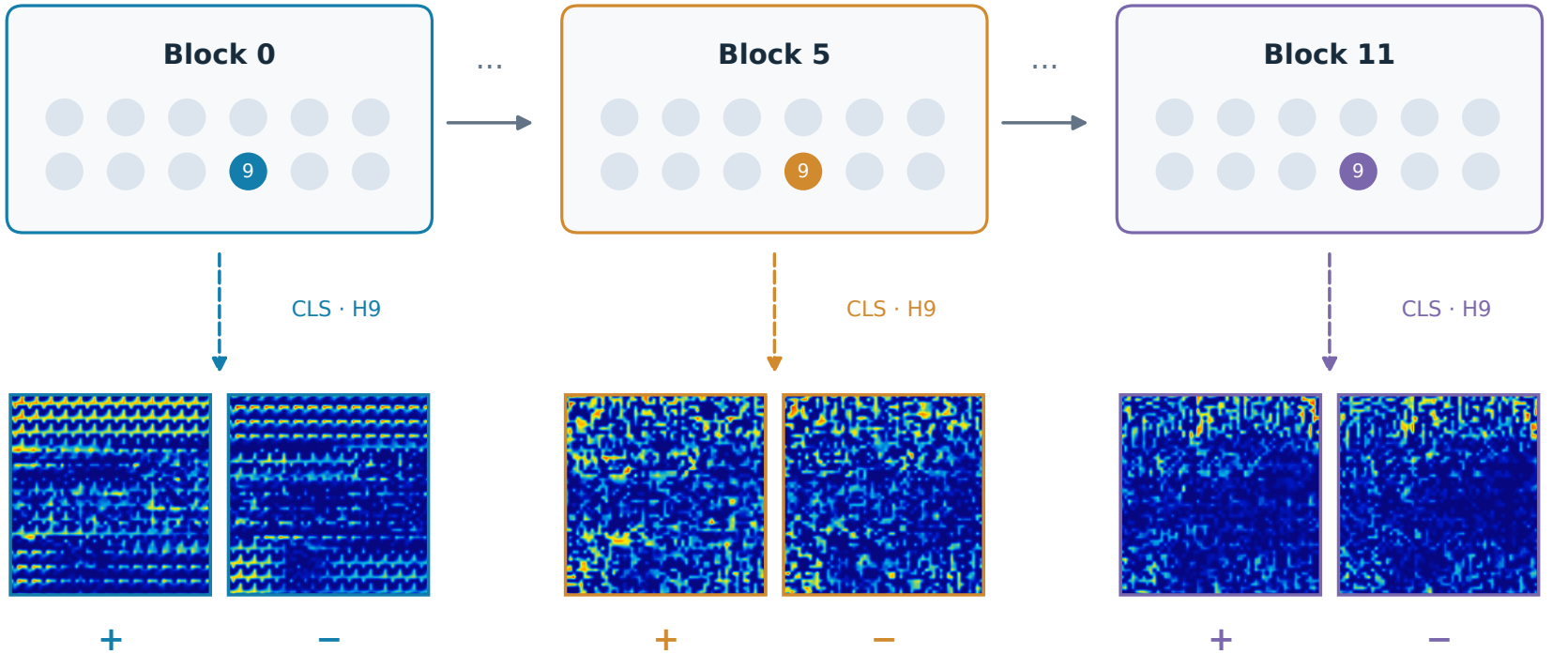
What is an NNS image in a ViT?

An input-space contribution map paired with a chosen internal response.

ViT computation



Patch tokens + CLS



NNS image

Same input coordinates

Choose a response

$$f_{\ell,h}(z) = u^\top \sum_j A_{\ell,h}(x)_{\mathrm{CLS},j} V_{\ell,h}(z)_j$$

Trace each input coordinate along the reference-to-image path

$$R_i = (x_i - x_{0i}) \int_0^1 \partial_i f_{\ell,h}(x_0 + \alpha(x - x_0))\, d\alpha$$